

definition of bad times. Preferences are also affected by ethics, psychological tendencies, peer decisions, and other social factors. We return to some of these utility functions in section 4.

3. Mean-Variance Utility

With mean-variance utility, asset owners care only about means (which they like), and variances (which they dislike). Mean-variance utility defines bad times as low means and high variances.

Mean-variance utility is given by:

$$U = E(r_p) - \frac{\gamma}{2} \text{var}(r_p), \quad (2.4)$$

where r_p is the return of the investor's portfolio and γ is her coefficient of risk aversion.

Mean-variance utility is closely related to CRRA utility (see equation (2.1)). In fact, we can consider expected utility using CRRA utility and mean-variance to be approximately the same, and we will do so for many purposes in this book.¹⁴ Thus, the γ in equation (2.4) has the same interpretation as the risk aversion in section 2.3, and we expect most investors' γ s to lie between 1 and 10. The parameter of $\frac{1}{2}$ in equation (2.4) is simply a scaling parameter; it is of no consequence and can be ignored.

Mean-variance utility is the workhorse utility of the investment industry. It was introduced by Harry Markowitz in his pathbreaking work in 1952, but Markowitz did not fully rationalize using mean-variance utility in an expected utility framework until 1979 in a paper co-authored with Haim Levy. Levy and Markowitz approximate any expected utility function by a mean-variance utility function:

$$E[U(1 + r_p)] \approx U(1 + E(r_p)) + \frac{1}{2} U''(1 + E(r_p)) \text{var}(r_p), \quad (2.5)$$

where $U''(\bullet)$ denotes the second derivative of the utility function. Utility functions are concave, so their second derivatives are negative and the second term in equation (2.5) is negative. Hence, equation (2.5) shows that an investor maximizing expected utility is approximately the same as an investor maximizing mean for a given level of variance, and thus equation (2.5) takes the same form as mean-variance utility in equation (2.4).

¹⁴ The portfolio choices made with CRRA utility and mean-variance utility are very close and will converge to each other under certain conditions, like employing (log) normal distributions and sampling intervals that tend to zero.